

Guideline Title: Narrow & Wide Complex Tachycardia

Guideline Number: GUID-3203-C006

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Successful treatment of the rapid heart rate by stabilizing the abnormal heart rhythm through therapeutic interventions.
- II. **DEFINITIONS:**
- III. **DEPARTMENT GUIDELINE:**
 - a. Assess and manage airway, breathing, and circulation. Apply oxygen if hypoxemic.
 - b. Apply cardiac monitor and confirm rhythm.
 - c. Obtain 12 lead ECG if possible.
 - d. Apply hands-free defibrillator pads to patient's bare chest.
 - e. If patient demonstrates rate-related cardiovascular compromise with signs and symptoms such as hypotension SBP < 90 mmHg, acutely altered mental status, shock, severe ischemic chest discomfort, acute heart failure/pulmonary edema occur at any point, or other signs of shock suspected to be due to a tachyarrhythmia, consider proceeding directly to **Cardioversion** per GUID3203-PR004.
 - i. Have the patient perform vagal maneuvers while preparing to cardiovert.
 - ii. Set J dose based on current recommended dosing.
 1. For narrow rhythm: 50-100 J
 2. For narrow irregular rhythm: 120-200 J
 3. For wide regular complex: 100 J
 4. For wide irregular: Defibrillation dose, not synchronized
 5. If no response after initial cardioversion, give 2nd dose of 200 J. Subsequent doses should be higher than initial dose up to max joule dose of monitor. (Other facility/ agency monitors maximum may be up to 360 J).
 - iii. Consider sedation or analgesia per GUID3203-G005 *Analgesia and Sedation Management* if MAP >65 mmHg.
 - iv. Obtain 12 lead ECG after cardioversion as soon as practical.
 - f. Establish IV/IO access (preferably large bore catheter).

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- g. Consider causes and correct before administration of adenosine (pain, dehydration, medication withdrawal, etc.).
- h. Consider Medication Administration for heart rate > 150 bpm with symptoms but stable.
- i. **Narrow Complex:** If the patient is stable with a regular QRS complex < 0.12 MS
 - i. Attempt vagal maneuvers.
 - ii. **Adenosine 6 mg rapid IV/IO** push followed by a 20 ml bolus of 0.9% NS.
 - 1. Use caution in patients with history of asthma or chronic obstructive pulmonary disease, particularly if they are actively bronchospastic as adenosine may cause bronchospasm.
 - 2. Initial dose may be reduced to 3mg in patients taking dipyridamole or carbamazepine or who have a transplanted heart.
 - iii. If the rate does not convert within 1-2 minutes, administer **Adenosine 12 mg rapid IV/IO** push followed by a 20 ml bolus of 0.9% NS.
 - iv. If Adenosine is not effective, consider administering beta-blocker or calcium channel blocker:
 - 1. **Metoprolol 5 mg IV/IO** push over 2 minutes. May repeat x 2 every 5 minutes until a total of 15 mg. Obtain an optimal heart rate of < 100 bpm. **Avoid with patients in CHF, heart block, valvular failure, or cocaine use**
 - OR-
 - 2. **Diltiazem 0.25 mg/kg IV/IO** push over 2 minutes. If not effective in 15 minutes, a second dose of 0.35 mg/kg IV/IO may be given.
 - a. If persistent tachycardia above 120 bpm after two boluses, begin **Diltiazem infusion at 5mg/h** and titrate to 15 mg/hr to maintain a HR of < 120 bpm.
- j. **Wide Complex:** If the patient is stable with a regular QRS complex > 0.12 ms
 - i. Consider **Adenosine 6 mg rapid IV/IO** push followed by a 20 ml bolus of 0.9% NS if the rhythm is regular and monomorphic.
 - ii. If adenosine is not effective or indicated, consider antiarrhythmic infusions:
 - 1. **Amiodarone 150 mg IV** bolus over 10 minutes followed by **Amiodarone infusion at 1 mg/min**. Amiodarone is the preferred initial agent for wide complex tachycardia -OR-
 - 2. **Procainamide infusion at 20-50 mg/min** until arrhythmia is suppressed, hypotension ensues, QRS duration increases by 50% or maximum dose of 17 mg/kg is administered. Follow with **Procainamide infusion at 1-4 mg/min** if rhythm is suppressed. Avoid in patients with prolonged QT. -OR-
 - 3. **Lidocaine 0.5-1.5 mg/kg IV/IO** bolus may repeat every 5-10 minutes at 0.5-0.75 mg/kg (max. dose 3 mg/kg). Follow with **Lidocaine infusion at 1-2 mg/min**, titrate up to 4 mg/min for effect if rhythm is suppressed.
 - iii. Consider **Magnesium sulfate 1-2 g IV/IO** over 5-20 minutes for Torsades de Pointes or known hypomagnesemia (chronic alcohol/diuretic use).
 - iv. If hyperkalemia is suspected, consider administering

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1. **Calcium Chloride 1 gram IV/IO** over 10 minutes.
 2. **Sodium Bicarbonate 8.4% 50 mEq IV/IO.**
 - k. Always consider the latest recommendations from the American Heart Association.
- IV. **DOCUMENTATION:**
- a. EMS Charts
- V. **REFERENCES:** American Heart Association, 2015.